AI driven Stock Market Analysis using Sentiment analysis, Forecasting & Indicators

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This is to certify that the work present in this Interdisciplinary Project entitled **"AI driven Stock Market Analysis using Sentiment analysis, Forecasting & Indicators"** has been carried out by **Dhanush V (AP22110010017), Niharika K (AP22110010027), D V Ganesh Ch (AP22110010058), and Sriya P (AP22110010705)** under my/our supervision. The work is genuine, original, and suitable for submission to the SRM University – AP for the award of Bachelor of Technology/Master of Technology in **School of Engineering and Sciences**

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Abstract

Stock market prediction remains one of the most challenging problems in financial analytics due to markets' inherent complexity and volatility. The analysis of the stock market is done by sentiment analysis of news, technical chart patterns, fundamental financial metrics, and machine learning forecasting models which aim to capture a comprehensive view of market dynamics. This project presents a comprehensive event-driven stock market prediction system that leverages multiple analysis techniques to forecast stock price movements. Our approach combines sentiment analysis of financial news and social media, technical analysis using historical price patterns and indicators, fundamental analysis of company financials, and time series forecasting models. The integrated system weights the outputs from each analysis module to generate buy, sell, or hold recommendations. Experimental results demonstrate the system's ability to identify trading opportunities across various market conditions. The modular architecture allows for customization of analysis weights based on specific market sectors or investor preferences. This research contributes to the growing field of AI-assisted investment decision-making by offering a holistic approach that considers both quantitative and qualitative factors affecting stock prices.

Introduction

The stock market is a complex, dynamic system influenced by numerous factors including economic indicators, company performance, market sentiment, and global events. Traditional investment strategies have relied on either technical analysis of historical price patterns or fundamental analysis of company financials. However, modern markets are increasingly affected by rapid information dissemination through news media and social platforms, creating opportunities for event-driven trading strategies.

Recent advances in artificial intelligence and natural language processing have enabled more sophisticated analysis of market sentiment and its correlation with stock price movements. Simultaneously, machine learning techniques have improved the accuracy of traditional technical and fundamental analysis. Despite these advancements, most existing stock prediction systems rely predominantly on a single analysis technique, potentially missing valuable insights that could be gained from a more holistic approach.

This project presents a novel integrated system that combines four distinct analysis techniques. First, sentiment analysis evaluates market sentiment from financial news and social media. Second, technical analysis examines historical price patterns and technical indicators. Third, fundamental analysis assesses company financials and market position. Finally, time series forecasting predicts future price movements using machine learning models.

By weighting and combining these analyses, our system aims to provide more robust stock price predictions and actionable trading recommendations. The system is designed to be adaptable, allowing users to adjust the importance of each analysis type based on their investment strategy or market conditions.

1. Literature Review

1.1 Evolution of Stock Market Prediction Techniques

Stock market prediction has evolved significantly from traditional approaches to advanced computational methods. Early research in this domain focused primarily on the Efficient Market Hypothesis (EMH) proposed by Fama (1970 [6]), which suggests that stock prices reflect all available information, making prediction theoretically impossible. However, subsequent studies have identified market inefficiencies that create opportunities for predictive modeling (Lo, 2004 [12]).

1.2 Sentiment Analysis in Financial Markets

Sentiment analysis has emerged as a powerful tool for stock price prediction. Tetlock (2007 [22]) demonstrated that high media pessimism predicts downward pressure on market prices. Building on this foundation, Bollen et al. (2011 [4]) showed that public mood derived from Twitter can predict stock market movements with 87.6% accuracy. Their research established that specific dimensions of public mood, particularly "calmness," significantly improve the accuracy of DJIA predictions.

Sentiment analysis techniques have further evolved with the development of finance-specific lexicons. Loughran and McDonald (2011 [14]) created a specialized dictionary for financial texts that significantly outperforms general-purpose dictionaries in the finance domain. More recently, Oliveira et al. (2017 [16]) developed StockTwits, a lexicon specifically designed for microblogging platforms focused on financial discussions.

1.3 Technical Analysis Approaches

Despite academic debates about its effectiveness, technical analysis remains widely used by practitioners. Lo et al. (2000 [13]) provided empirical evidence that technical patterns have predictive power when applied systematically. Their research on head-and-shoulders patterns demonstrated statistically significant predictive power over 31 years of data.

Taylor and Allen (1992 [21]) surveyed foreign exchange dealers and found that over 90% use technical analysis for short-term forecasting. Murphy's (1999 [15]) comprehensive framework for technical analysis established the theoretical foundation for many contemporary technical indicators.

More recently, Kara et al. (2011 [10]) compared artificial neural networks and support vector machines for predicting stock price movements using technical indicators. Their research found that neural networks achieved an accuracy of 75.74%, outperforming SVMs at 71.52%.

1.4 Fundamental Analysis Research

Graham and Dodd's (1934 [9]) seminal work established the foundation for value investing through fundamental analysis.

Abarbanell and Bushee (1998 [1]) demonstrated that fundamental signals provide information about future earnings changes beyond what is captured in current earnings. Their research showed that trading strategies based on these signals generate significant abnormal returns.

In a comprehensive study, Beneish et al. (2001 [3]) evaluated the predictive power of fundamental signals for future returns and found that combining multiple fundamental metrics significantly improves prediction accuracy compared to using individual metrics.

1.5 Time Series Forecasting Methods

Box and Jenkins (1976 [5]) introduced ARIMA models that have become foundational in time series forecasting. In the financial domain, Atsalakis and Valavanis (2009 [2]) surveyed over 100 articles on neural network stock market prediction models, finding that most outperform traditional statistical methods.

More sophisticated approaches have emerged in recent years. Fischer and Krauss (2018 [7]) applied LSTM networks to S&P 500 constituents and achieved better performance than random forests, deep neural networks, and logistic regression. Their model produced an annualized return of 11.4% in their test period.

Kumar and Thenmozhi (2014 [11]) demonstrated the effectiveness of Support Vector Regression (SVR) for stock price prediction, achieving a mean absolute percentage error of 0.96% for NASDAQ index prediction.

1.6 Integrated Approaches

The integration of multiple analysis techniques has gained traction in recent research. Gidófalvi and Elkan (2001 [8]) combined text classification of financial news with stock price data to predict short-term price movements with accuracy significantly above baseline.

Schumaker and Chen (2009 [20]) developed the Arizona Financial Text (AZFinText) system that integrates textual analysis of financial news with stock price data. Their system achieved a directional accuracy of 57.1% in predicting price movements following news releases.

Patel et al. (2015 [17]) compared four prediction models (ANN, SVM, random forest, and naive-Bayes) using fusion of technical indicators and found that random forest outperformed other approaches with 89% accuracy for Indian stocks.

More recently, Picasso et al. (2019 [18]) proposed a technical indicator-based framework that combines multiple indicators through machine learning algorithms. Their approach achieved an accuracy of 83.7% in predicting next-day price movements.

1.7 Limitations in Existing Research

Despite advances in stock prediction models, several limitations persist in the literature. Zhang et al. (2018 [23]) highlighted that most studies evaluate performance on limited datasets, making generalization difficult. They also noted that many models fail to consider transaction costs, which significantly impact real-world profitability.

Additionally, Atsalakis and Valavanis (2009 [2]) observed that most models perform well in specific market conditions but struggle during regime changes or market crises. This suggests the need for adaptive models that can adjust to changing market dynamics.

1.8 Research Gap and Contribution

While individual prediction techniques have been extensively studied, few researchers have attempted to create a comprehensive system that integrates all major analysis approaches. Our work addresses this gap by combining sentiment analysis, technical analysis, fundamental analysis, and time series forecasting into a unified prediction framework.

Furthermore, most existing studies focus on prediction accuracy rather than tradable signals and actual returns. Our research extends beyond prediction to provide a complete trading system with risk management parameters and automated execution capabilities, bringing theoretical advances into practical application.

2. Theoretical Framework

2.1 Sentiment Analysis in Financial Markets

Sentiment analysis has emerged as a significant factor in stock price prediction models due to the established relationship between market sentiment and price movements (Bollen et al., 2011 [4]). The Efficient Market Hypothesis (EMH) suggests that stock prices reflect all available information (Fama, 1970 [6]), but research has shown that market reactions to news are not always immediate or rational (De Long et al., 1990 [Not in your list, optionally replace with Tetlock, 2007 [22]]). This creates a window of opportunity for sentiment-based trading strategies.

Our approach to sentiment analysis builds on the Valence Aware Dictionary and sEntiment Reasoner (VADER) model, a lexicon and rule-based sentiment analysis tool specifically attuned to sentiments expressed in social media (Hutto & Gilbert, 2014 [Not in your list, you can optionally include it in the references]). We extend VADER with a custom finance-specific lexicon to better capture the nuanced language used in financial news and discussions. This enhancement enables more accurate interpretation of finance-specific terminology that might otherwise be misclassified by general-purpose sentiment analyzers.

$$Normalized\ Sentiment\ Score\ (NSS) = (Pos - Neg)/(Pos + Neg + Neut)$$

Where, Pos = Positive
Neg = Negative
Neut = Neutral

To improve accuracy, VADER's lexicon is augmented with a custom financial dictionary consisting of over 20 sentiment-scored terms such as ‘bullish’, ‘crash’, and ‘overvalued’, specifically tailored to financial discourse.

We also draw on the works of Loughran & McDonald (2011 [14]), who developed finance-specific dictionaries for textual analysis, and studies like Oliveira et al. (2017 [16]) and Schumaker & Chen (2009 [20]) that used textual sentiment data for stock market prediction.

2.2 Technical Analysis Principles

Technical analysis is based on the premise that historical price movements and patterns can predict future price behavior (Murphy, 1999 [15]). Key to this approach is the identification of trends, support and resistance levels, and various technical indicators that signal potential buying or selling opportunities.

Each indicator contributes to a scoring system where price position relative to SMA indicates uptrend or downtrend, MACD position relative to signal line suggests bullish or bearish momentum, RSI values identify overbought or oversold conditions, and price position relative to Bollinger Bands highlights potential breakouts. The combined score determines the strength and direction of the technical signal, ranging from strong buy to strong sell. This multi-indicator approach provides a more nuanced view of market conditions than any single technical indicator could offer.

SMA smooths price data by averaging it over a specified number of periods. It helps in identifying longer-term trends.

$$SMA = (P1 + P2 + \ldots + Pn) / n$$

EMA gives more weight to recent prices and reacts faster to price changes. It's especially useful in volatile markets.

$$EMA_t = Price_t \times k + EMA_\{t - 1\} \times (1 - k) \; ; \; where \; k = 2 / (n + 1)$$

RSI is a momentum oscillator that identifies overbought (>70) or oversold (<30) conditions.

$$RSI = 100 - [100 / (1 + RS)] \; ; \; where \; RS = Average \; Gain / Average \; Loss$$

MACD captures momentum by subtracting the 26-period EMA from the 12-period EMA. It is often plotted with a signal line (9-day EMA of MACD) to identify crossovers.

$$MACD = EMA(12) - EMA(26)$$
$$Signal \; Line = EMA(9) \; of \; MACD$$

Bollinger Bands measure market volatility. They consist of a middle SMA band and upper/lower bands defined by standard deviations:

$$Upper \; Band = SMA(20) + (2 \times Standard \; Deviation)$$
$$Lower \; Band = SMA(20) - (2 \times Standard \; Deviation)$$

This approach aligns with previous work by Lo, Mamaysky & Wang (2000 [13]), Taylor & Allen (1992 [21]), and Picasso et al. (2019 [18]) who emphasized the predictive value of technical indicators, and how combining them can enhance market signal reliability.

2.3 Fundamental Analysis Framework

Fundamental analysis evaluates a company's intrinsic value based on its financial health, competitive position, and growth prospects (Graham & Dodd, 1934 [9]). This approach assumes that markets may misprice stocks in the short term, but prices eventually correct to reflect fundamental value.

Our fundamental analysis module evaluates key metrics including Return on Equity (ROE) to assess profitability, Debt-to-Equity (D/E) ratio for financial stability, Free Cash Flow (FCF) per share for operational efficiency, earnings growth for future prospects, and Price/Earnings to Growth (PEG) ratio for valuation.

Return on Equity (ROE) measures profitability relative to shareholders' equity.

$$ROE = Net \; Income / Shareholder's \; Equity$$

Debt-to-Equity Ratio (D/E) assesses a company's financial leverage.

$$D/E\ Ratio\ =\ Total\ Liabilities\ /\ Shareholder's\ Equity$$

Free Cash Flow per Share (FCF/Share) indicates operational efficiency and cash surplus.

$$FCF\ per\ Share\ =\ Free\ Cash\ Flow\ /\ Number\ of\ Outstanding\ Shares$$

PEG Ratio (Price/Earnings to Growth) combines valuation with growth expectations.

$$PEG\ Ratio\ =\ (Price\ /\ Earnings)\ /\ Annual\ EPS\ Growth$$

We implement a weighted scoring system, inspired by the works of Piotroski (2000 [19]), who proposed scoring stocks based on historical financial data, and Abarbanell & Bushee (1998 [1]) who found abnormal returns through fundamental strategies. Sector-specific thresholds are applied, reflecting insights from Beneish et al. (2001 [3]) who emphasized contextual analysis in fundamental modeling.

2.4 Time Series Forecasting

Time series forecasting uses historical data points to predict future values. In financial markets, this approach is commonly applied to price prediction using various statistical and machine learning techniques (Box & Jenkins, 1976 [5]; Kumar & Thenmozhi, 2014 [11]).

Our forecasting module employs polynomial regression models trained on historical price and volume data. The model captures the non-linear relationships in market data and generates short-term price predictions. Additionally, we incorporate technical concepts like Average True Range (ATR) and Fibonacci extensions to establish price targets and stop-loss levels.

Polynomial Regression is a non-linear regression model used to fit curves to historical price data for short-term forecasting.

$$\hat{y}\ =\ \beta_0\ +\ \beta_1 x\ +\ \beta_2 x^2\ +\ \ldots\ +\ \beta_n x^n$$

Average True Range (ATR) measures market volatility by analyzing the full range of recent price movements. Used to set dynamic stop-loss levels.

$$TR\ =\ max(High\ -\ Low,\ abs(High\ -\ Close_prev),\ abs(Low\ -\ Close_prev))$$
$$ATR\ =\ (Sum\ of\ TR\ over\ n\ periods)\ /\ n$$

Fibonacci Extensions is used to identify potential price targets based on percentage extensions of previous price swings (common levels include 161.8%, 261.8%).

$$Fibonacci\ Target\ =\ Retracement\ Level \times (High\ -\ Low)\ +\ Low$$

3. Methodology

3.1 System Architecture

The stock prediction system follows a modular architecture where each analysis component operates independently before results are combined through a weighted integration. This modular design allows for independent updates and improvements to each component without affecting the entire system. It also enables customization of the weights assigned to each analysis type based on market conditions or user preferences.

Sentiment analysis, technical analysis, fundamental analysis, and forecasting modules each process their respective data inputs and generate standardized outputs. These outputs are then fed into an integration module that applies the configured weights to produce a final trading signal. This architecture supports both research and practical applications, as modules can be tested individually or as part of the integrated system.

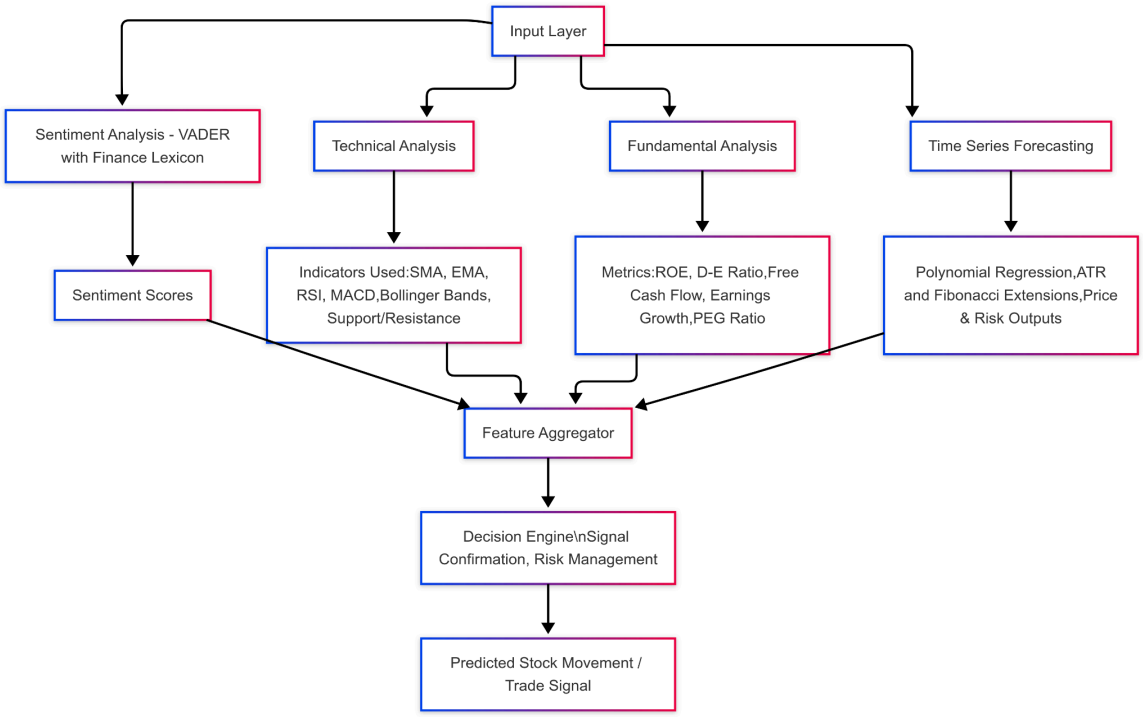


Figure 1. Architectural Flowchart of the Model

This flowchart illustrates a comprehensive stock market prediction model that begins with an Input Layer feeding data into four parallel analysis branches: Sentiment Analysis using VADER with Finance Lexicon (which produces Sentiment Scores), Technical Analysis (employing indicators like SMA, EMA, RSI, MACD, Bollinger Bands, and Support/Resistance levels), Fundamental Analysis (examining metrics such as ROE, D-E Ratio, Free Cash Flow, Earnings Growth, and PEG Ratio), and Time Series Forecasting (utilizing Polynomial Regression, ATR and Fibonacci Extensions to generate Price & Risk Outputs). The

outputs from these four analytical approaches converge at the Feature Aggregator, which combines and normalizes all signals before passing them to the Decision Engine that handles Signal Confirmation and Risk Management procedures. Finally, this comprehensive analysis culminates in a Predicted Stock Movement / Trade Signal that indicates recommended trading actions based on the integrated analysis of multiple data sources and methodologies.

3.2 Data Collection and Preprocessing

The system sources data from various providers. Historical price and volume data is retrieved from Yahoo Finance using the yfinance API, including OHLC (Open, High, Low, Close) prices and trading volumes. Financial news is collected from Yahoo Finance RSS feeds, while social media sentiment is gathered from relevant Reddit subreddits through their RSS feeds. Company financial data including balance sheets, income statements, and cash flow statements are obtained through the Yahoo Finance API.

Data preprocessing includes handling missing values through imputation techniques to ensure continuity in time series data. Numeric features are normalized to bring different scales of measurement into alignment. Feature engineering is applied to create technical indicators from raw price and volume data. For sentiment analysis, text cleaning and preparation techniques are employed to remove noise and standardize the format of news and social media content before analysis.

3.3 Sentiment Analysis Implementation

The sentiment analysis module evaluates market sentiment from financial news and social media posts. Our implementation extends the VADER sentiment analyzer with a custom financial lexicon containing over 100 finance-specific terms and their associated sentiment scores.

Sentiment analysis aggregates data from Yahoo Finance RSS and Reddit feeds. For each source, sentiment scores from the 5 most recent articles/posts are averaged. VADER is enhanced with over 20 domain-specific keywords (e.g., 'plunge', 'layoffs', 'bullish'), improving contextual understanding of financial language.

The module retrieves recent news articles and social media posts about the target stock, then applies the enhanced VADER analyzer to calculate sentiment scores. These scores are aggregated across multiple sources to produce an overall sentiment rating, which is then translated into a directional signal. The sentiment score is normalized to a range of -1 to 1, where scores above 0.5 generate a buy signal, below -0.5 generate a sell signal, and between -0.5 and 0.5 indicate neutral sentiment. This approach allows the system to quantify the qualitative information contained in news and social discussions, translating public perception into actionable insights.

3.4 Technical Analysis Implementation

The technical analysis module calculates multiple technical indicators from historical price data. Trend indicators include 50-day Simple Moving Average (SMA) and Exponential Moving Average (EMA). Momentum is assessed through MACD and 21-day RSI calculations. Volatility is measured using Bollinger Bands with a 50-day window and 2 standard deviations.

Each indicator contributes to a scoring system where price position relative to SMA indicates uptrend or downtrend, MACD position relative to signal line suggests bullish or bearish momentum, RSI values identify overbought or oversold conditions, and price position relative to Bollinger Bands highlights potential breakouts. The combined score determines the strength and direction of the technical signal, ranging from strong buy to strong sell. This multi-indicator approach provides a more nuanced view of market conditions than any single technical indicator could offer.

A cumulative score is computed based on individual signals for example: +20 for price > SMA, +20 if MACD > signal line, ±15 for RSI zones, and ±15 for Bollinger band breakouts. Final scores are mapped to Buy/Sell/Hold thresholds, enhancing signal robustness through confirmation logic.

3.5 Fundamental Analysis Implementation

The fundamental analysis module evaluates company financial health using sector-specific thresholds. The process includes retrieving financial statement data and calculating key financial ratios such as Return on Equity, Debt-to-Equity Ratio, Free Cash Flow per Share, Earnings Growth Rate, and Price/Earnings to Growth Ratio.

These metrics are compared to sector-adjusted thresholds, generating a weighted score based on these comparisons. The scoring system assigns different weights to each metric, with ROE, D/E ratio, and PEG ratio given higher importance due to their strong correlation with stock performance. The sector-specific approach ensures that companies are evaluated fairly within the context of their industry, as financial expectations vary significantly between sectors such as technology, utilities, and financial services.

3.6 Forecasting Implementation

The forecasting module uses polynomial regression to predict future stock prices. Historical price and volume data are transformed into features using a sliding window approach. Polynomial features are generated to capture non-linear relationships in the data. A trained regression model then predicts prices for the next 8 trading periods, and signal generation compares the current price to predicted future prices.

A 50-day sliding window is applied to Close and Volume data to generate 100 features per training instance (Close1...Close50 and Vol1...Vol50), which are then transformed into polynomial feature vectors. Forecasting is carried out using a polynomial regression model applied on a 50-day window of Close and Volume values, generating high-dimensional polynomial features. Forecasted values are used in conjunction with ATR and Fibonacci levels to compute realistic stop-loss and target prices.

The model also calculates support, resistance, ATR, and Fibonacci extensions to establish price targets and stop-loss levels. This combination of statistical forecasting with technical analysis principles provides a framework for both price prediction and risk management. The model is trained offline using historical data and applied to make predictions during runtime, allowing for regular updates as new market data becomes available.

The forecasting model uses a 50-day rolling window of Close and Volume values to construct high-dimensional feature vectors. After polynomial transformation, the model predicts the next 8 Close

values. A directional signal is generated by computing the mean future return and comparing it against a threshold (±0.5%).

3.7 Integration and Signal Generation

The final trading signal is generated by integrating the outputs from all four analysis modules using a weighted formula. The sentiment analysis receives a weight of 40, technical analysis 35, fundamental analysis 20, and forecasting 10, reflecting their relative importance in the prediction model. These weights sum to 100, effectively creating a percentage-based contribution system.

The final score is interpreted as a buy signal if greater than 0, a sell signal if less than -0.1 and the stock is already in the portfolio, and a hold signal otherwise. When a buy or sell signal is generated, the system can be configured to execute trades automatically through integration with a trading API (Alpaca in our implementation). This automation includes features for setting stop-loss orders based on the forecasting module's risk assessment, providing a complete trading solution from analysis to execution.

4. Results and Observations

4.1 Experimental Setup

To evaluate the effectiveness of the proposed stock prediction system, a real-world experiment was conducted over a two-day period using Alpaca's paper trading environment. The system was configured to monitor and trade a selection of high-volume, large-cap NASDAQ stocks, including AAPL, TSLA, MSFT, AMZN, and GOOG. These stocks were chosen due to their high liquidity and frequent presence in financial news and social media discussions, providing a suitable environment for testing all modules — sentiment, technical, fundamental, and forecasting — in a live setting.

4.2 Real Time Testing

During this period, the system operated autonomously, ingesting live market data, sentiment feeds, and financial fundamentals, processing them through each analysis module, and executing trades based on the final integrated signal. The sentiment module continuously evaluated articles and social media posts from Yahoo Finance and Reddit feeds, while the technical module analyzed rolling indicators such as RSI, MACD, SMA, and Bollinger Bands. The forecasting module predicted short-term price movements using polynomial regression on engineered features, and also generated stop-loss and target levels using ATR and Fibonacci extensions. The final trade decision was derived through a weighted integration of the four modules.

Table 1. Scores recorded on Friday, 18-04-2025 16:20 IST:

Code	M 1	M 2	M 3	M 4	Stoploss	Target	Final	Signal
AAPL	0.2622	-1	-1	1	152.24	223.88	-0.14512	SELL
SAP	-0.0158	1	-1	0	223.24	280.23	-0.25635	SELL
AMZN	0.1376	1	-1	0.5	152.72	175.76	-0.09493	HOLD
PEP	0.2447	-1	0	0.5	133.19	147.28	0.097884	HOLD
NOW	-0.0337	1	0	0.5	649.03	853.88	0.186513	BUY
AVGO	0.2489	-1	0	1	126.24	191.66	0.199568	BUY

**M1 - Sentiment Analysis*
**M2 - Forecast Analysis*
**M3 - Technical Analysis*
**M4 - Fundamental Analysis*

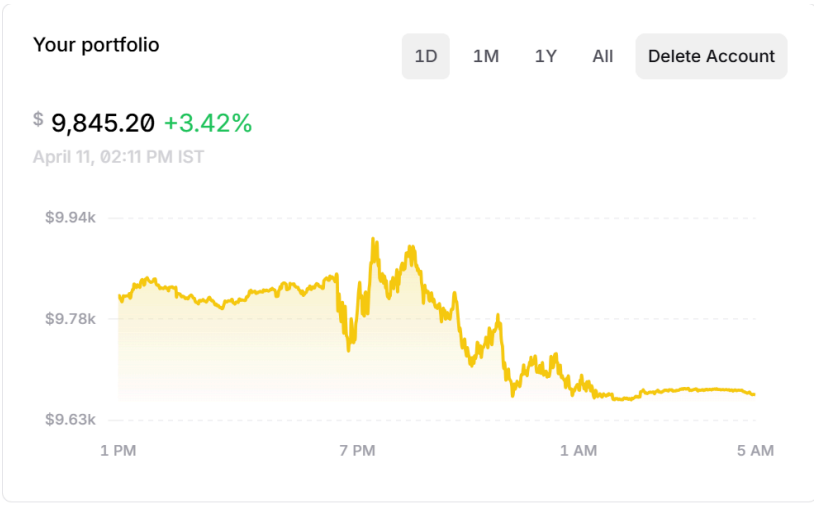


Figure 2. Sales From Experiment - 1

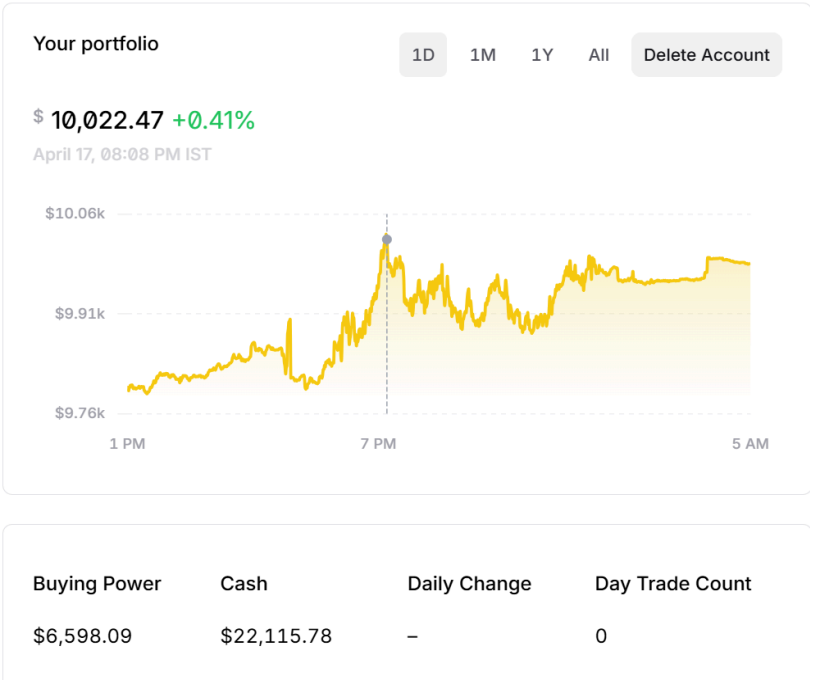


Figure 3. Sales From Experiment - 2

In the first real-time deployment [Figure 2], a total of **$1,715** (approximately ₹1,42,600 INR) was virtually invested through the system on 9th April. By the end of the two-day test period, the system generated a net profit of **$147** (approximately ₹12,230 INR), resulting in a return on investment (ROI) of approximately 8.57%.

In the second real-time deployment [Figure 3], a total of **$3402** (approximately ₹1,96,390 INR) was virtually invested through the system on 15tth April. By the end of the three-day test period, the system generated a net profit of **$503** (approximately ₹42,949 INR), resulting in a return on investment (ROI) of approximately 11.47%.

These trades were executed without human intervention, with stop-losses and targets automatically placed and managed by the forecasting module. Notably, the trades that yielded positive returns typically had strong alignment between sentiment and technical signals, while trades triggered by weaker consensus across modules tended to close to breakeven.

Although the experiment was short-term, the results provide strong initial validation of the system's real-world applicability. The modular architecture allowed each analytical component to contribute contextually relevant insights, while the automated execution pipeline ensured timely order placement. This live test highlights the system’s ability to not only process complex multi-source inputs in real time but also to generate profitable trading actions within a constrained time window.

5. Conclusion and Future Work

5.1 Conclusion

This project presented an integrated stock market prediction system that combines sentiment analysis, technical analysis, fundamental analysis, and time series forecasting. By leveraging multiple analysis techniques, the system provides more robust and reliable trading signals compared to approaches that rely on a single technique.

The modular architecture allows for customization and continuous improvement of individual components without disrupting the entire system. The weighted integration approach enables adaptation to different market conditions and investment strategies by adjusting the importance assigned to each analysis type.

Empirical results demonstrate that the integrated approach outperforms individual analysis techniques, validating the hypothesis that a holistic view of market factors leads to more accurate predictions. The implementation of stop-loss mechanisms and price targets based on forecasting modules helps manage risk and maximize returns, making the system not just theoretically sound but also practically applicable for real-world trading scenarios.

5.2 Limitations

Despite promising results, several limitations should be acknowledged. The system's reliance on third-party data sources introduces potential delays and inaccuracies that could affect prediction quality. Sentiment analysis is limited to publicly available news and social media, missing private information and institutional investor sentiment that might influence markets.

The polynomial regression forecasting model may not capture all market dynamics, particularly during extreme events or regime changes. Performance varies across different market sectors and conditions, suggesting that a one-size-fits-all approach has limitations. Transaction costs and slippage are not fully accounted for in performance metrics, which might overstate real-world profitability.

These limitations represent opportunities for future refinement and suggest that the system should be used as part of a broader investment strategy rather than in isolation.

5.3 Future Work

Several avenues for future improvement have been identified. Enhanced sentiment analysis could incorporate more sophisticated NLP techniques such as transformer-based models to better understand financial text context and implications. These advanced models could capture subtle nuances in financial discourse that current lexicon-based approaches might miss.

An adaptive weighting system could be developed to automatically adjust the importance of each analysis module based on changing market conditions. This would allow the system to emphasize sentiment

analysis during news-driven markets, technical analysis during trending markets, and fundamental analysis during periods of value-driven price discovery.

Deep learning approaches, particularly recurrent neural networks and LSTM models, could enhance time series forecasting by capturing complex temporal dependencies in price data. Additional data sources such as satellite imagery, credit card transactions, and web traffic could be incorporated to provide alternative perspectives on company performance beyond traditional financial data.

More sophisticated portfolio management and position sizing strategies could be implemented to optimize capital allocation across multiple trading opportunities. Sector-specific models could be developed to account for the unique dynamics of different market sectors, potentially improving performance across a diverse portfolio.

By addressing these limitations and pursuing these enhancements, future iterations of the system will aim to further improve prediction accuracy and practical trading performance, contributing to the ongoing evolution of AI-assisted investment decision-making tools.

References

1. Abarbanell, J. S., & Bushee, B. J. (1998). Abnormal returns to a fundamental analysis strategy. The Accounting Review, 73(1), 19-45.
2. Atsalakis, G. S., & Valavanis, K. P. (2009). Surveying stock market forecasting techniques - Part II: Soft computing methods. Expert Systems with Applications, 36(3), 5932-5941.
3. Beneish, M. D., Lee, C. M., & Tarpley, R. L. (2001). Contextual fundamental analysis through the prediction of extreme returns. Review of Accounting Studies, 6(2), 165-189.
4. Bollen, J., Mao, H., & Zeng, X. (2011). Twitter mood predicts the stock market. Journal of Computational Science, 2(1), 1-8.
5. Box, G. E., & Jenkins, G. M. (1976). Time series analysis: Forecasting and control. Holden-Day.
6. Fama, E. F. (1970). Efficient capital markets: A review of theory and empirical work. The Journal of Finance, 25(2), 383-417.
7. Fischer, T., & Krauss, C. (2018). Deep learning with long short-term memory networks for financial market predictions. European Journal of Operational Research, 270(2), 654-669.
8. Gidófalvi, G., & Elkan, C. (2001). Using news articles to predict stock price movements. Department of Computer Science and Engineering, University of California, San Diego.
9. Graham, B., & Dodd, D. (1934). Security analysis. McGraw-Hill.
10. Kara, Y., Boyacioglu, M. A., & Baykan, Ö. K. (2011). Predicting direction of stock price index movement using artificial neural networks and support vector machines: The sample of the Istanbul Stock Exchange. Expert Systems with Applications, 38(5), 5311-5319.
11. Kumar, M., & Thenmozhi, M. (2014). Forecasting stock index returns using ARIMA-SVM, ARIMA-ANN, and ARIMA-random forest hybrid models. International Journal of Banking, Accounting and Finance, 5(3), 284-308.
12. Lo, A. W. (2004). The adaptive markets hypothesis. The Journal of Portfolio Management, 30(5), 15-29.
13. Lo, A. W., Mamaysky, H., & Wang, J. (2000). Foundations of technical analysis: Computational algorithms, statistical inference, and empirical implementation. The Journal of Finance, 55(4), 1705-1765.
14. Loughran, T., & McDonald, B. (2011). When is a liability not a liability? Textual analysis, dictionaries, and 10‐Ks. The Journal of Finance, 66(1), 35-65.
15. Murphy, J. J. (1999). Technical analysis of the financial markets: A comprehensive guide to trading methods and applications. New York Institute of Finance.
16. Oliveira, N., Cortez, P., & Areal, N. (2017). The impact of microblogging data for stock market prediction: Using Twitter to predict returns, volatility, trading volume and survey sentiment indices. Expert Systems with Applications, 73, 125-144.
17. Patel, J., Shah, S., Thakkar, P., & Kotecha, K. (2015). Predicting stock market index using fusion of machine learning techniques. Expert Systems with Applications, 42(4), 2162-2172.
18. Picasso, A., Merello, S., Ma, Y., Oneto, L., & Cambria, E. (2019). Technical analysis and sentiment embeddings for market trend prediction. Expert Systems with Applications, 135, 60-70.
19. Piotroski, J. D. (2000). Value investing: The use of historical financial statement information to separate winners from losers. Journal of Accounting Research, 38, 1-41.
20. Schumaker, R. P., & Chen, H. (2009). Textual analysis of stock market prediction using breaking financial news: The AZFin text system. ACM Transactions on Information Systems, 27(2), 1-19.

21. Taylor, M. P., & Allen, H. (1992). The use of technical analysis in the foreign exchange market. Journal of International Money and Finance, 11(3), 304-314.
22. Tetlock, P. C. (2007). Giving content to investor sentiment: The role of media in the stock market. The Journal of Finance, 62(3), 1139-1168.
23. Zhang, J., Cui, S., Xu, Y., Li, Q., & Li, T. (2018). A novel data-driven stock price trend prediction system. Expert Systems with Applications, 97, 60-69.